

Gravitational Lensing by Torsion Stars: Layered Foci beyond the Schwarzschild Deflection

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October 2, 2026

Abstract

Double spacetime theory recovers the exterior of a spherical mass as a Schwarzschild chart and freezes the redshift at a quasi-horizon r_* . On both the three-axis ceiling, where $1 < r_*/r_s < 1.005$, and the one-axis saturation, the photon sphere lies outside r_* , so the critical impact parameter remains $b_c = 3\sqrt{3}GM/c^2$. Rays with $b > b_c$ never meet a node. Under the equal-scale ansatz each interval of π contains one node, with $1/6 < r_2/r_1 < 4/(5\pi)$, and the plateaux are exponentially quiet. Reading each node as a thin signed shell, after $|J| \leq 3\pi^2/8$ has cut off the pole of $1/\gamma_s$, yields a weak-deflection map. A point mass has one tangential critical curve. On a thin shell the tangential ratio stays finite at the limb, while the radial slope does not. An attractive shell, with b_m the Einstein radius of its mass and R its areal radius, has an interior radial critical curve once $R > b_m/\sqrt{2}$, and an interior tangential curve only for $b_m/\sqrt{2} < R < b_m$, with that tangential curve outside the radial one. A repulsive shell has neither curve on its disk. The nodal ratio is the ratio of the nodes, not of these curves. The masses are unfixed, a photon has vanishing stiffness, and $F = k/(\gamma_s r^2)$ is not the null geodesic, so magnifications are not predicted. The solar delay, galactic Einstein rings, and the emission rings of M87* and Sgr A* meet the exterior chart, because those rays do not enter r_* . The interval the chart adds is the round trip of a radial null ray from the photon sphere to r_* : $20.96 GM/c^3$ on the three-axis ceiling and $11.44 GM/c^3$ on the one-axis saturation, on either side of the photon half-orbit $16.32 GM/c^3$. A late echo train is not that return.

1 Introduction

The double spacetime theory refuses a shared continuum. Each massive particle carries a usual Minkowski frame and a dual frame, and gravity is the torsional mismatch of that pair [1]. On the exterior of a spherical mass the mismatch is read as a radial boost, and the boost reproduces the Schwarzschild coframe. Inside a sufficient density the same mismatch changes sign with radius. The parent theory calls the resulting object a torsion star: a layered condensate with no event horizon and no central singularity. The layers are not an extra assumption pasted onto the Schwarzschild interior. Under the equal-scale ansatz they are the zeros of a single interference factor, one node in each interval of π .

Light answers a different question from the fall of a massive particle. The written particle action gives a photon vanishing stiffness, so the action does not accelerate a photon's rapidities. What deflects light, on the exterior, is the null geometry of the coframe. What is asked in this note is how that geometry, together with the nodal barriers, maps a source direction to image directions. The ordinary Schwarzschild lens has one critical curve in the weak field, the Einstein ring of the total mass, and one critical impact parameter in the strong field, the photon sphere. A thin shell can do more than that single ring, but the further curves are not the limb. The divergence at the limb sits in the radial slope. Where a curve lies is decided by comparing the shell's areal radius with the Einstein radius of the mass it carries.

Two statements have to be kept apart from the start. The exterior deflection, the location of the photon sphere relative to the quasi-horizon, and the window on neighboring nodal radii are readings of results already obtained in the parent theory. The identification of a node with a thin signed shell, and the lens map built from the projected mass of that shell, are a model. The model is the weak-deflection reading of the barriers. It is not a completed null integration through a strong-field interior, and it does not assign a mass to each barrier.

The note follows that division. Section 2 records the chart, the quasi-horizon, and the nodal window. Section 3 recalls the Schwarzschild lens. Section 4 separates rays that turn outside r_* from rays that enter it. Section 5 constructs the layered map. Section 6 locates the critical curves. Section 7 compares the two readings with the Shapiro delay, with galactic rings, and with the emission rings of M87* and

Sgr A*. Section 8 isolates the intervals on which the frozen chart and general relativity can actually be told apart.

2 The chart inherited from the parent theory

On the exterior Schwarzschild domain the radial boost rapidity and the redshift factor are

$$\varphi(r) = -\frac{1}{2} \log(1 - r_s/r), \quad A(r) = 1 - \frac{r_s}{r}, \quad r_s = \frac{2GM}{c^2}, \quad (1)$$

with $r > r_s$. The coframe $(\sqrt{A}, A^{-1/2}, r, r \sin \theta)$ induces the Schwarzschild metric. Three quadratic quantities live on this chart and are not the same object: the finite-angle mismatch $J = \frac{1}{2}\varphi^2$, the field seed $J_{\text{field}} = \frac{1}{2}(\varphi')^2$, and the teleparallel scalar T . They agree at leading order in the weak field and nowhere else. The Einstein equation belongs to T after a tetrad variation. It is not an identity for J .

Classical Schwarzschild sends $\varphi \rightarrow \infty$ and $A \rightarrow 0$ as $r \rightarrow r_s^+$. The admissible cone cuts that divergence. Identifying the three-axis ceiling $|J| = 3\pi^2/8$ with a radial pure boost inverts to

$$\varphi_{\max} = \frac{\pi\sqrt{3}}{2}, \quad A_{\min} = e^{-\pi\sqrt{3}}, \quad \frac{r_*}{r_s} = \frac{1}{1 - A_{\min}}. \quad (2)$$

The parent theory records the window $0 < A_{\min} < 10^{-2}$ and $1 < r_*/r_s < 1.005$. For $r \leq r_*$ the chart freezes $\varphi = \varphi_{\max}$ and $A = A_{\min}$. The frozen factor is strictly positive. A null surface $A = 0$ does not form. The same parent passage flags the identification as a three-axis ceiling read onto one boost axis. A one-axis admissible boost saturates at $\varphi = \pi/2$, where $A = e^{-\pi}$ and $r_*/r_s = 1/(1 - e^{-\pi})$. That ratio is still less than $3/2$.

The layered interior is organized by a different scalar. Under the equal-scale ansatz the interference factor, written in $x = \ell/(2r)$, is

$$\gamma_s(x) = \cosh x \cos x - \sinh x \sin x, \quad (3)$$

and its derivative collapses to

$$\gamma'_s(x) = -2 \cosh x \sin x. \quad (4)$$

Each interval $[n\pi, (n+1)\pi]$ therefore contains exactly one node, and the sign of γ_s alternates from plateau to plateau. The n -th node lies in

$$n\pi + \frac{\pi}{4} < x_n < n\pi + \frac{\pi}{2}, \quad (5)$$

and the first two areal radii satisfy

$$\frac{1}{6} < \frac{r_2}{r_1} < \frac{4}{5\pi} \approx 0.255. \quad (6)$$

The ratio depends only on the dimensionless roots. The characteristic length ℓ sets the absolute scale and cancels from every ratio. Plateaux are screened by $1/\cosh(\ell/(2r))$. What grows toward the centre is the barrier at the node, where γ_s has a simple zero and $1/\gamma_s$ diverges in the equal-scale idealization.

That divergence is not a physical column density. On the admissible cone $|J| \leq 3\pi^2/8$, and the parent dictionary confines the dimensionless teleparallel density to $-8 < r^2 T < 27$. The spike in the effective source is finite. The cone does not convert the window into a unique surface density. Section 5 replaces each finite spike by a thin shell of some mass m_n , and leaves m_n unfixed.

One further distinction belongs here because the lens map will otherwise be misread. In the written particle action a photon is a particle of vanishing stiffness. Vanishing stiffness removes the acceleration of the common rapidity. It does not force the photon onto the vacuum, and it does not make $F = k/(\gamma_s r^2)$ the force on a ray. The symbol F is the effective force in the massive layer reading. Null deflection on the exterior is the geodesic of the coframe (1). Null deflection through the barriers is the model of Section 5.

3 The Schwarzschild lens

A static spherical line element

$$ds^2 = -A(r) c^2 dt^2 + A(r)^{-1} dr^2 + r^2 d\Omega^2$$

conserves an energy E and an angular momentum L . Their ratio is the asymptotic impact parameter $b = L/E$. Radial turning points of a null ray obey

$$b^2 = \frac{r_0^2}{A(r_0)}. \quad (7)$$

Circular photon orbits are the critical points of A/r^2 . For the exterior factor (1),

$$\frac{d}{dr} \left(\frac{A}{r^2} \right) = -\frac{2}{r^3} + \frac{3r_s}{r^4},$$

so the only positive root is the photon sphere

$$r_{\text{ph}} = \frac{3}{2} r_s = \frac{3GM}{c^2}. \quad (8)$$

The corresponding impact parameter is

$$b_c = \frac{r_{\text{ph}}}{\sqrt{A(r_{\text{ph}})}} = 3\sqrt{3} \frac{GM}{c^2} = \frac{3\sqrt{3}}{2} r_s. \quad (9)$$

At that root A/r^2 has a maximum, so the circular orbit is unstable. Rays from infinity with $b > b_c$ turn at some $r_0 > r_{\text{ph}}$. Rays with $b < b_c$ pass r_{ph} .

The azimuthal advance from a turning point $r_0 > r_{\text{ph}}$ out to infinity and back is

$$\Delta\varphi(b) = 2 \int_{r_0}^{\infty} \frac{dr}{r^2 \sqrt{b^{-2} - A(r)/r^2}}, \quad \alpha_{\text{ext}}(b) = \Delta\varphi(b) - \pi, \quad (10)$$

with b and r_0 tied by (7). In the weak field, $b \gg r_s$, the integral reduces to

$$\alpha_{\text{ext}}(b) = \frac{4GM}{c^2 b} + O\left(\left(\frac{r_s}{b}\right)^2\right). \quad (11)$$

The leading term is the quantity tested by light bending at the solar limb. The same Schwarzschild factor fixes the Shapiro delay. In the parametrized post-Newtonian writing both the bending and the delay scale with $(1 + \gamma)/2$, and the chart (1) is the case $\gamma = 1$.

For a thin lens at angular-diameter distance D_L from the observer, with a source at D_S and a lens–source distance D_{LS} , the lens equation relates the angular source position β to the angular image position θ by

$$\beta = \theta - \bar{\alpha}(\theta), \quad \bar{\alpha}(\theta) = \frac{D_{LS}}{D_S} \alpha(D_L \theta). \quad (12)$$

The Einstein angle of a point mass M is defined by

$$\theta_E^2 = \frac{4GM}{c^2} \frac{D_{LS}}{D_L D_S}, \quad (13)$$

so that the weak point-mass deflection becomes $\bar{\alpha}(\theta) = \theta_E^2/\theta$. Two eigenvalues of the lens map vanish on different sets [2]. The tangential eigenvalue vanishes where

$$\bar{\alpha}(\theta) = \theta, \quad (14)$$

and the radial eigenvalue vanishes where

$$\frac{d\bar{\alpha}}{d\theta} = 1. \quad (15)$$

For the point mass, (14) has the single positive root $\theta = \theta_E$, the Einstein ring of a source on the optic axis. The radial derivative is $-\theta_E^2/\theta^2$, which does not equal 1 for any real θ . A point mass therefore has one tangential critical curve and no radial critical curve. That is the lens a torsion star must exceed if its layers are to draw further foci.

4 Two classes of rays

Join the exterior factor (1) for $r > r_*$ to the frozen value $A = A_{\min}$ for $r \leq r_*$. The join is continuous, because r_* was defined by $A(r_*) = A_{\min}$. On the frozen interval

$$\frac{d}{dr} \left(\frac{A}{r^2} \right) = -\frac{2A_{\min}}{r^3} < 0.$$

The frozen redshift contributes no critical point. The only critical point of the joined function A/r^2 on $r > 0$ remains r_{ph} , provided $r_{\text{ph}} > r_*$.

Section 2 already places both ceilings below the photon sphere: $r_*/r_s < 1.005$ on the three-axis window, and $r_*/r_s = 1/(1 - e^{-\pi}) < 3/2$ on the one-axis saturation. In either reading,

$$r_{\text{ph}} > r_*. \quad (16)$$

The critical curve of rays from infinity is fixed in the exterior chart, at the impact parameter (9). Layers inside r_* do not move it.

The same comparison splits the rays into two classes.

A ray of the exterior class has $b > b_c$. Its turning point lies at $r_0 > r_{\text{ph}} > r_*$. The whole of the deflected path is in the Schwarzschild domain, and the deflection is (10). No node is encountered. Every weak-field test whose impact parameter is large compared with r_s , and every strong-field test whose observable is the angular diameter $2b_c/D_A$ of this critical curve, belongs to this class.

A ray of the penetrating class has $b < b_c$. It passes the photon sphere, enters $r < r_*$, and, on the frozen chart alone, reaches a periapsis where (7) gives

$$r_{\text{peri}} = b\sqrt{A_{\min}}. \quad (17)$$

The periapsis is positive for every finite b . The ray returns. There is no horizon to capture it. Because the maximum of the effective potential still lies below the energy of any ray with $b < b_c$, the returning ray can recross r_{ph} and reach infinity. Near b_c the passage past the unstable orbit can wind many times. That winding is the photon-ring sequence of the exterior geometry. It is one critical curve, visited repeatedly. It is not a sequence of nodal limbs.

Equation (17) is chart geometry. It does not know about γ_s . A sphere of areal radius r inside the freeze-out is grazed by the ray whose periapsis lies on that sphere, and the asymptotic impact parameter of that ray is $r/\sqrt{A_{\min}}$. The map is homogeneous of degree one, so neighboring nodal spheres are grazed at impact parameters in the ratio of their areal radii. The graze is accessible to a ray from infinity only when that impact parameter lies below b_c , that is when $r < b_c\sqrt{A_{\min}}$. An outer node near r_* is crossed by every penetrating ray and is not a periapsis limb on the sky. The shell model of the next section is a second, weak-deflection reading of the same nodes. It is not a correction added on top of (10) for the same ray.

5 A layered lens map

The plateaux of γ_s are exponentially quiet, so the smooth column they contribute is neglected. Each node is a barrier. The equal-scale zero of γ_s would make $1/\gamma_s$ non-integrable across the node. The admissible window of Section 2 replaces that pole by a finite spike. The model of this section collapses the spike further, to a thin spherical shell of areal radius R and mass m . The mass m may be negative. A positive shell is the idealization of an attractive barrier, a negative shell of a repulsive one. Nothing in the cone fixes the value of m , nor the fraction of the total mass that sits on any one node.

Let $\sigma = m/(4\pi R^2)$ be the surface density. A line of sight at impact parameter $b < R$ meets the shell twice. The angle between the line of sight and the radial normal has cosine $\sqrt{R^2 - b^2}/R$, so the projected column is

$$\Sigma(b) = \frac{2\sigma R}{\sqrt{R^2 - b^2}} \quad (b < R), \quad (18)$$

and $\Sigma(b) = 0$ for $b > R$. The column diverges at the limb $b = R$. That divergence is the ideal thin limit. A spike of finite width, as required by the cone, keeps Σ finite and peaked at the limb.

The mass projected inside a cylinder of radius b is

$$M_{\text{cyl}}(b) = \int_0^{\min(b, R)} 2\pi b' \Sigma(b') db'.$$

For $b \leq R$ the integral is elementary,

$$M_{\text{cyl}}(b) = 4\pi\sigma R \int_0^b \frac{b' db'}{\sqrt{R^2 - b'^2}} = \frac{m}{R} \left(R - \sqrt{R^2 - b^2} \right), \quad (19)$$

and for $b \geq R$ one has $M_{\text{cyl}}(b) = m$. The weak-field deflection, with the general-relativistic factor $4G/c^2$ rather than the Newtonian $2G/c^2$, is

$$\alpha(b) = \frac{4G M_{\text{cyl}}(b)}{c^2 b}. \quad (20)$$

Outside the shell this is the point-mass law. On the axis, $R - \sqrt{R^2 - b^2} \sim b^2/(2R)$, so $\alpha \rightarrow 0$ as $b \rightarrow 0$, as spherical symmetry requires. The slope does not stay bounded at the limb: $dM_{\text{cyl}}/db = (m/R)b/\sqrt{R^2 - b^2}$ diverges as $b \rightarrow R^-$. A finite thickness replaces the divergence by a steep but finite peak.

A torsion star, in this reading, is a stack of such shells, one at each nodal radius $R_n = r_n$, with signed masses m_n . The cylindrical masses add. For the weak-field normalization one requires

$$\sum_n m_n = M, \quad (21)$$

so that a ray with b larger than every node sees the point-mass deflection of the total mass. That normalization matches (11). It is a choice of the model. It is not a theorem that the barriers carry the whole mass, and it is not the deflection law of an exterior-class ray. Exterior-class rays are already accounted for by (10), which knows the total mass through r_s and does not pass through the shells.

The sign of m_n alternates, because the sign of γ_s alternates from plateau to plateau and each barrier is the transition between them. Consequently $\alpha(b)$ need not be monotone in b . A monotone deflection, of the point-mass form θ_E^2/θ , produces two images of a source off the axis. A non-monotone deflection can produce more. Those further images are the complex foci of the layered lens. They exist in this model when the corresponding critical curves exist, which is the question of the next section.

6 Critical curves

Substitute (20) into the lens equation (12). With the Einstein angle (13) of the total mass, and with $b = D_L \theta$,

$$\bar{\alpha}(\theta) = \frac{\theta_E^2}{\theta} \frac{M_{\text{cyl}}(D_L \theta)}{M}, \quad (22)$$

where M_{cyl} now denotes the sum of the shell contributions. Outside every node, $M_{\text{cyl}} = M$ by (21), and $\bar{\alpha}$ collapses to the point-mass deflection. The tangential condition (14) still has the root $\theta = \theta_E$, and the radial condition (15) still has none. The Einstein ring of the total mass survives the layering. It is the critical curve of rays that stay outside every shell.

Inside one shell the two conditions part company, and they do not part company by pinning a critical curve to the limb.

The ideal column (18) diverges at $b = R$. The mass enclosed by the cylinder does not. Let b_m be the Einstein radius of the mass on the shell, and set $c = (b_m/R)^2$, negative when that mass is negative. Inside the shell the reduced deflection depends on $\xi = b/R$ through

$$\frac{\bar{\alpha}}{\theta} = \frac{c}{1 + \sqrt{1 - \xi^2}}, \quad \frac{d\bar{\alpha}}{d\theta} = \frac{c}{\sqrt{1 - \xi^2} (1 + \sqrt{1 - \xi^2})}. \quad (23)$$

On the axis the shear vanishes, and both expressions equal $c/2$. The tangential ratio only doubles on the way out, and it meets the finite value c at the limb. The radial slope does not stay finite. It runs to $+\infty$ when the shell is attractive and to $-\infty$ when the shell is repulsive. What diverges at the limb is the radial eigenvalue. The tangential eigenvalue remains the comparison of a finite ratio with unity.

For an attractive shell the two comparisons fall into a window measured by b_m . The radial slope equals one at exactly one radius inside the shell when $R > b_m/\sqrt{2}$. The tangential ratio equals one inside the shell when

$$\frac{b_m}{\sqrt{2}} < R < b_m, \quad (24)$$

and in that window the tangential curve lies outside the radial one. Outside the shell the map is the point mass of the shell's own mass: an Einstein ring whenever $R < b_m$, and never a radial critical curve. A

shell inside the window therefore carries both interior curves and the exterior ring. Only the ring remains when the shell is smaller than $b_m/\sqrt{2}$. Only the interior radial curve remains when the shell is larger than b_m . The tangential curve meets the node only at $R = b_m$, the outer edge of the window, where it has migrated out to the limb.

A repulsive shell keeps both $\bar{\alpha}/\theta$ and the interior slope negative, so its disk contains neither critical curve. Outside, it is a negative point mass. There is still no Einstein ring, and there is a radial curve at radius b_m only when the shell itself lies inside that radius.

The cone replaces the ideal limb by a spike of finite width. The infinite radial slope becomes a steep peak, and a narrow attractive spike can still carry the radial eigenvalue through zero close to the limb. Smoothing does not move the tangential curve onto the limb. That curve remains where the enclosed-mass ratio equals one.

The mass on the shell is unfixed, so the window is a criterion and not a count of images. The nodal ratio (6) remains the ratio of the nodes. Neighboring critical curves are not obliged to stand in it. In a stack, mass already enclosed adds the point-mass deflection of that interior, finite at the next limb. The sign of the radial divergence is still the sign of the shell being crossed, while the tangential ratio at the limb is the Einstein ratio of everything enclosed there, not of the local shell alone. A repulsive shell wrapped around a heavier interior can therefore still satisfy $\bar{\alpha} = \theta$. The alternation of γ_s does not alternate the type of the critical curve from node to node.

The frozen chart assigns the nodal ratio to another family, the periapsis grazes. Section 4 gives a nodal sphere the grazing impact parameter $b_n = r_n/\sqrt{A_{\min}}$ when $b_n < b_c$. The redshift factor cancels in b_{n+1}/b_n , so the angular ratio of the grazes is again (6). A graze is not a weak-deflection critical curve. The critical curve sits on the node only at the boundary $R = b_m$, and even there the two maps differ in scale by $1/\sqrt{A_{\min}}$. Neither reading fixes m_n . Which grazes can be reached from infinity is settled in Section 8.

The photon-ring sequence is not the nodal ratio. Successive half-orbits of the single unstable curve at b_c are demagnified, for the Schwarzschild geometry, by a factor $e^{-\pi}$ per half-orbit, and their image-plane radii converge to that one curve [8]. A pair of nodes is a pair of distinct areal radii. An exponential pile-up at b_c , a pair whose ratio lies in (6), and a pair of thin-shell critical curves fixed by (24) are three arrangements of light on the sky. The present note does not integrate the ray that both winds at r_{ph} and crosses the barriers. It separates the arrangements and leaves the combined deflection uncomputed.

7 Comparison with observations

7.1 The exterior class

The solar tests sit in the exterior class. A ray that grazes the photosphere has an impact parameter of order the solar radius, larger than r_s by a factor of order 10^5 , and it does not enter a nucleus. Nuclear nodes, whatever their femtometer scale, are not on the path. The deflection is (11) evaluated at the solar limb, and the delay is the Shapiro delay of the same factor A . Cassini tracking of radio links past the Sun gives

$$\gamma = 1 + (2.1 \pm 2.3) \times 10^{-5} \quad (25)$$

[3]. The chart is the case $\gamma = 1$. The measurement constrains any correction that would rescale the exterior bending and delay together. It does not constrain the interior shells, which that ray does not meet.

Galactic strong lensing is the same class in a weaker field. The Einstein angle (13) and the angular size r_*/D_L of the quasi-horizon are related by $\theta_E^2 \sim (r_s/D_L)(D_{LS}/D_S)$. The ratio of the two angles is then the square root of an angle measured in radians, and for any galactic lens that square root is many orders of magnitude below unity. Structure at r_* does not move the Einstein ring, the image positions, or the time delay at the precision of those observations. A double Einstein ring formed by two sources at two redshifts is two exterior-class rings. It is not a pair of nodal limbs. Flux-ratio anomalies among images of a single quasar are likewise an exterior-class question, read in the literature as substructure in the lens, and this note does not assign them to torsional nodes.

7.2 The critical curve of M87* and of Sgr A*

The emission ring of M87* has diameter $42 \pm 3 \mu\text{as}$, surrounding a central depression whose flux ratio is $\gtrsim 10:1$. The angular gravitational radius adopted with that ring is $\theta_g = GM/(c^2 D) = 3.8 \pm 0.4 \mu\text{as}$, at

distance $16.8 \pm 0.8 \text{ Mpc}$, and the corresponding mass is $(6.5 \pm 0.7) \times 10^9 M_\odot$ [4]. The critical curve (9) has angular diameter

$$\frac{2b_c}{D_A} = 6\sqrt{3}\theta_g. \quad (26)$$

Since $6\sqrt{3} \approx 10.392$,

$$6\sqrt{3} \times (3.8 \pm 0.4) \mu\text{as}$$

is about $39.5 \pm 4.2 \mu\text{as}$. That interval overlaps $42 \pm 3 \mu\text{as}$. The overlap is the comparison the exterior chart supports. The published ring is lensed emission around the critical curve, not the critical curve itself, and the mass inference already uses a shadow scaling of that order. Agreement at this precision is a check that the photon sphere lies where (16) places it. It is not a detection of a horizon. The horizon is what the quasi-horizon replaces, and the critical curve does not consult the replacement.

Sgr A* is the same comparison at a second mass. The emission ring has diameter $51.8 \pm 2.3 \mu\text{as}$ [5]. Calibrated to general-relativistic simulations, the angular gravitational radius is $4.8^{+1.4}_{-0.7} \mu\text{as}$ [6]. The central value of (26) is about $49.9 \mu\text{as}$. The uncertainty on θ_g runs from 4.1 to $6.2 \mu\text{as}$, and $6\sqrt{3}$ times that interval runs from about 43 to $64 \mu\text{as}$. The measured ring lies inside the interval. Again the exterior critical curve is consistent with the image, at the precision of the published gravitational radius. Both images are single rings. What that agreement leaves undecided is Section 8.

8 Where the predictions differ

The agreements in Section 7 do not separate the joined chart from the Schwarzschild lens. The rays that produce them stay outside $r_\star$. Equation (16) puts the photon sphere in the exterior, so $2b_c/D_A$ is the Schwarzschild diameter. The Cassini delay and the galactic rings have impact parameters far outside r_s . The ratio (6) is a ratio of nodes, and of the periaapsis grazes of Section 4. The thin-shell critical curves of Section 6 are not obliged to repeat it. On Earth baselines the next-generation array expects to separate at most the $n = 0$ and $n = 1$ subrings of the single curve b_c , and that separation already requires super-resolution [9]. It does not measure a ratio bounded by $4/(5\pi)$.

The brightness inside the shadow does not separate them either. A penetrating ray returns, by (17), so the interior need not be empty, and an optically thick plasma can keep it dark. The M87* contrast $\gtrsim 10:1$ limits the sum. A surface that thermalizes the incident power, or that reflects it perfectly, is a further object, and the Sgr A* spectrum and image disfavor those two [7]. The freeze-out $A = A_{\min}$ is neither of them, so the bound does not transfer to $r_\star$.

What the joined chart does fix is a Killing time. On the exterior, a radial null ray satisfies $c dt = dr/A$. The round trip from the photon sphere down to the quasi-horizon and back is therefore

$$\Delta t = \frac{2}{c} \int_{r_\star}^{r_{\text{ph}}} \frac{dr}{A(r)} = \frac{2}{c} \left[r + r_s \ln |r - r_s| \right]_{r_\star}^{r_{\text{ph}}}. \quad (27)$$

General relativity has no such return: a radial ray that passes r_{ph} with $b = 0$ reaches r_s in infinite Killing time and does not come back. The integral is finite once $r_\star > r_s$.

The value depends on which ceiling fixes $r_\star$. For the three-axis factor $A_{\min} = e^{-\pi\sqrt{3}}$ one finds

$$\Delta t = 20.96 \frac{GM}{c^3}. \quad (28)$$

For the one-axis saturation $A = e^{-\pi}$ the quasi-horizon sits farther above r_s , the logarithm is shorter, and

$$\Delta t = 11.44 \frac{GM}{c^3}. \quad (29)$$

The parent theory flags the three-axis reading as a ceiling transplanted onto one axis. Until that identification is settled, the chart predicts an interval in $\{11.44, 20.96\} GM/c^3$, not a single number.

The natural clock already present in the exterior is the half-period of the unstable photon orbit. With $c = 1$ and $M = GM/c^2$, the angular velocity on $r = 3M$ is $d\varphi/dt = 1/(3\sqrt{3}M)$, so

$$T_{1/2} = 3\pi\sqrt{3} \frac{GM}{c^3} \approx 16.32 \frac{GM}{c^3}. \quad (30)$$

The two returns fall on either side of this half-period, by $4.6 GM/c^3$ and by $4.9 GM/c^3$. Successive photon-ring images of a single flash are separated by $T_{1/2}$ in the Schwarzschild geometry [8]. A return from $r_\star$ is

a different interval, but not a different order of magnitude. For a mass $4 \times 10^6 M_\odot$, $GM/c^3 \approx 20$ s, and the three intervals are about 4, 5, and 7 minutes. For the M87* mass quoted in Section 7 they are a few days. A flare campaign that resolves a delay of several GM/c^3 can in principle tell the return from one photon half-orbit. No such separation is in hand.

The same interval shows why a late gravitational-wave echo is not the prediction. Echo searches bound reflecting surfaces whose proper distance to the horizon is tiny, and they look for a pulse train long after the ringdown. The round trip (27) is one ringdown cycle, not that train. A non-detection of a late train is what a cavity as wide as $r_* - r_s$ leads one to expect. It does not confirm the cavity. Confirming it would require the ringdown itself to be shifted by a reflector whose coefficient is known. The joined chart supplies the metric coefficient A . It does not supply a boundary condition for a gravitational wave, and the shell masses m_n remain unfixed. The parent remark that echoes are already hinted in the data is not a result of this calculation.

Periapsis limbs remain a second, weaker target. They are accessible from infinity only for areal radii $r < b_c \sqrt{A_{\min}}$. On the three-axis ceiling that cut is $0.171 r_s$. Nodes outside it are crossed by every penetrating ray and do not appear as separate grazes. Nodes inside it would be grazed at impact parameters in the areal ratio, which is again (6), but only after the ceiling is chosen and only if the limb is bright enough to see. That is a deeper cut on the same unresolved ratio, not a new observable.

9 Conclusion

The exterior of a torsion star lenses light as the Schwarzschild chart lenses light. The photon sphere lies outside the quasi-horizon for both ceilings recorded in the parent theory, the critical impact parameter remains $b_c = 3\sqrt{3} GM/c^2$, and the Einstein ring of the total mass survives in the weak field. That is why the solar delay, the galactic rings, and the emission rings of M87* and Sgr A* meet the theory at the precision those measurements actually have.

Further foci require the barriers. The layered map of this note collapses each nodal spike, already bounded by the admissible cone, to a thin signed shell, and reads the lens from the projected mass. An isolated attractive shell adds an interior radial critical curve when its radius exceeds $b_m/\sqrt{2}$, and an interior tangential curve, outside that radial one, only for $b_m/\sqrt{2} < R < b_m$. A repulsive shell adds neither curve on its own disk. The nodal ratio $1/6 < r_2/r_1 < 4/(5\pi)$ remains the ratio of the nodes and of the accessible periapsis grazes. It is not the ratio of these critical curves. The mass on each shell is not determined, so the model does not predict which window is occupied, nor the magnifications, nor the deflection of a ray that both winds at the photon sphere and crosses the barriers.

The interval the chart itself adds is the round trip (27). General relativity does not return the ray. The three-axis and one-axis ceilings return it in $20.96 GM/c^3$ and $11.44 GM/c^3$, on either side of one photon half-orbit. That is the comparison a flare timed to a few GM/c^3 could make. A late echo train is a different comparison, and it is not the one this surface predicts.

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